

4 (a) charge / potential(ratio must be clear) B1 [1]

(b) potential (at surface of sphere) = $Q / 4\pi\epsilon_0 R$ M1
 $C = Q / V = 4\pi\epsilon_0 R$ A0 [1]

(c) (i) $C = 4\pi \times 8.85 \times 10^{-12} \times 0.63$ C1
 $= 7.0 \times 10^{-11}$ A1
farad / F B1 [3]

(ii) energy = $\frac{1}{2}CV^2$ C1
 $0.25 \times \frac{1}{2}C \times (1.2 \times 10^6)^2 = \frac{1}{2}CV^2$ C1
 $V = 6.0 \times 10^5 \text{ V}$ A1 [3]
(use of 0.75 rather than 0.25, allow max 2 marks)

[Total: 8]

5 (a) (i) concentric circles, anticlockwise (minimum 3 circles) M1